Coding & Solution

Date	1 July 2026
Team ID	SWTID-2026-9706
Project Name	Voice Based Concept understanding Analyser
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/Palak-web021/VBCUA
Programming Language(s)	Python
Framework(s) Used	Streamlit
Key Features Implemented	Audio upload & playback, Whisper speech-to-text, Sentence-BERT semantic similarity, Librosa audio feature extraction (filler ratio, pause ratio, energy), weighted scoring engine, waveform visualization, downloadable PDF report, session-based state (no login required)
Pending / Incomplete Features	Multi-language support, expanded reference concept bank, persistent storage across sessions
Setup / Run Instructions	1. Install ffmpeg (winget install ffmpeg) — required for Whisper's audio decoding 2. Create virtual environment: python -m venv vbcu_env 3. Activate it: vbcu_env\Scripts\activate 4. Install dependencies: pip install -r requirements.txt 5. Run the app: streamlit run app.py 6. Opens at localhost:8501

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Partial
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes